A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Chemical equilibria

At AS you predicted which way an equilibrium moves. Now you put numbers on it. The equilibrium constants K_c and K_p measure exactly where an equilibrium lies, you can calculate them from experimental data, and they explain why changing the temperature behaves so differently from changing concentration, pressure or adding a catalyst.

SPEC 13.1 Writing K_c for homogeneous and heterogeneous equilibria

WHAT THIS SPEC POINT ASKS

Deduce an expression for K_c, in terms of equilibrium concentrations, for homogeneous and heterogeneous systems.

The rule

For aA + bB ⇌ cC + dD (all in solution or all gases):

$$K_c = [C]^c[D]^d \div ([A]^a[B]^b)$$

- **Products on top**, reactants underneath.
- Each concentration is raised to the **power of its coefficient** in the balanced equation.
- **Square brackets** = equilibrium concentration in mol dm^{-3}.
- **Homogeneous**: everything in the same phase. **Heterogeneous**: more than one phase.
- In a heterogeneous system, **leave out pure solids** (and pure liquids, such as water used as the solvent). Their "concentration" is fixed, so it's already part of K.

Examples

Equilibrium	K_c
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	$[HI]^2 \div ([H_2][I_2])$
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	$[NH_3]^2 \div ([N_2][H_2]^3)$
$CH_3COOH(l) + C_2H_5OH(l) \rightleftharpoons CH_3COOC_2H_5(l) + H_2O(l)$	$[CH_3COOC_2H_5][H_2O] \div ([CH_3COOH][C_2H_5OH])$

Equilibrium	K_c
$Fe^{2+}(aq) + Ag^{+}(aq) \rightleftharpoons Fe^{3+}(aq) + Ag(s)$	$[Fe^{3+}] \div ([Fe^{2+}][Ag^{+}])$ (solid silver left out)
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$	$[CO_2]$

In the **ester** equilibrium water **is** included: it's a product mixed in with the other liquids, not a solvent present in huge excess.

Units of K_c

Substitute mol dm^{-3} for every concentration and cancel.

- $[HI]^2 \div ([H_2][I_2])$: $(mol\ dm^{-3})^2 \div (mol\ dm^{-3})^2 \rightarrow$ **no units**.
- $[NH_3]^2 \div ([N_2][H_2]^3)$: $(mol\ dm^{-3})^2 \div (mol\ dm^{-3})^4 = (mol\ dm^{-3})^{-2}$ = **$mol^{-2}\ dm^6$** ($dm^6\ mol^{-2}$).
- $[CO_2]$: **$mol\ dm^{-3}$**.

Units of Kc for $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

1. Write the expression: $K_c = \dfrac{[NH_3]^2}{[N_2][H_2]^3}$.
2. Substitute the unit: $\dfrac{(mol\,dm^{-3})^2}{(mol\,dm^{-3})(mol\,dm^{-3})^3} = \dfrac{(mol\,dm^{-3})^2}{(mol\,dm^{-3})^4}$.
3. Cancel: $(mol\,dm^{-3})^{-2} = mol^{-2}\,dm^6$.

A quick check: there are 2 molecules on top and 4 underneath, so the unit is (mol dm^{-3})$^{2-4}$.

MARK-SCHEME WORDING

- "$K_c = [NO]^2[Cl_2] \div [NOCl]^2$" (WCH14 Jun 2025 Q14). DO NOT AWARD round brackets, "p", or the 2 written inside the bracket.
- Heterogeneous: "$K_c = [CO_2]$" for a solid decomposing (WCH14 Oct 2025 Q14).
- Doubling every coefficient **squares** K (WCH14 Oct 2020 Q7).

Often asked as: write K_c (often as the first mark of a calculation); MCQ on K_c for a heterogeneous system; how K changes when the equation is reversed or multiplied.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Change the equation, change K: **reverse** it → 1/K; **double** it → K^2; **halve** it → $\sqrt{K}$ (WCH14 Jan 2025 Q5: halving the coefficients turned 4×10^{24} into 2.0×10^{12}).

COMMON MISTAKE

Including a solid. Writing the equation instead of the expression. Putting reactants on top.

SPEC 13.2 Writing K_p in terms of partial pressures

WHAT THIS SPEC POINT ASKS

Deduce an expression for K_p for homogeneous and heterogeneous systems, in terms of equilibrium partial pressures in atm.

Partial pressure

- In a mixture of gases, the **partial pressure** of a gas is the pressure it would exert if it filled the container on its own.
- **Mole fraction**: x_A = moles of A ÷ total moles of gas.
- **Partial pressure**: $p_A = x_A \times$ total pressure.
- The partial pressures **add up to the total pressure**, and the mole fractions add up to 1.

K_p expressions

Like K_c, but with **partial pressures** (written p_{NH_3} or $p(NH_3)$) and **only gases**:

Equilibrium	K_p	Units
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	$p(NH_3)^2 \div (p(N_2) \times p(H_2)^3)$	atm^{-2}
$\frac{1}{2}N_2(g) + 1\frac{1}{2}H_2(g) \rightleftharpoons NH_3(g)$	$p(NH_3) \div (p(N_2)^{\frac{1}{2}} \times p(H_2)^{1\frac{1}{2}})$	atm^{-1}
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$	$p(NO_2)^2 \div p(N_2O_4)$	atm
$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$	$p(SO_3)^2 \div (p(SO_2)^2 \times p(O_2))$	atm^{-1}

Equilibrium	K_p	Units
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$	$p(CO_2)$	atm
$C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$	$p(CO) \times p(H_2) \div p(H_2O)$	atm

- Solids and liquids are left out; gaseous **products** still count.
- Units come from cancelling atm, exactly as for K_c.

MARK-SCHEME WORDING

- "$K_p = p(NH_3) \div (p(N_2)^{½} \times p(H_2)^{1½})$" (WCH14 Jan 2026 Q15). Round brackets, "pp", "P" or "$p(NH_3)$" are all allowed.
- **DO NOT AWARD square brackets** in K_p (every mark scheme that asks says so), or an expression with no p at all.

Often asked as: write K_p (part of nearly every K_p calculation); give its units.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Fractional coefficients** give fractional powers and unusual units (atm^{-1} for the $½N_2 + 1½H_2$ equation) (WCH14 Jan 2026 Q15).

COMMON MISTAKE

Square brackets in K_p: examiner reports call this "disappointingly" common. Square brackets mean concentration.

Including a solid such as $CaCO_3$ in K_p.

SPEC 13.3 Calculating K_c and K_p, with units

WHAT THIS SPEC POINT ASKS

Calculate values of K_c and K_p, with units where needed, for homogeneous and heterogeneous reactions from experimental data.

Method (use an ICE table)

1. Write the balanced equation and the K expression.
2. **I**nitial moles; **C**hange (use the mole ratio); **E**quilibrium moles.
3. For K_c: divide each equilibrium amount by the **volume** (dm^3) to get concentrations.
   For K_p: find the **total equilibrium moles** of gas → mole fractions → partial pressures.
4. Substitute, calculate, and work out the units.

WORKED EXAMPLE

Worked example 1: K_c with no units (volume cancels)

0.50 mol CH_3COOH and 1.00 mol C_2H_5OH reach equilibrium; 0.423 mol of ethyl ethanoate has formed. Find K_c.

$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$

Initial: 0.50, 1.00, 0, 0

Change: −0.423, −0.423, +0.423, +0.423

Equilibrium: 0.077, 0.577, 0.423, 0.423

K_c = (0.423/V × 0.423/V) ÷ (0.077/V × 0.577/V): the V's **cancel**, so moles can be used.

K_c = 0.1789 ÷ 0.04443 = **4.0 (no units)**

WORKED EXAMPLE

Worked example 2: K_c with units

0.200 mol PCl_5 is heated in a 2.00 dm^3 flask: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. At equilibrium 0.120 mol PCl_5 remains.

Equilibrium moles: PCl_5 0.120; PCl_3 0.080; Cl_2 0.080

Concentrations: $[PCl_5]$ = 0.0600, $[PCl_3]$ = 0.0400, $[Cl_2]$ = 0.0400 mol dm^{-3}

K_c = (0.0400 × 0.0400) ÷ 0.0600 = **0.0267 mol dm^{-3}**

WORKED EXAMPLE

Worked example 3: K_p for the Haber process

1.00 mol N_2 and 3.00 mol H_2 reach equilibrium at a total pressure of 200 atm; 0.40 mol NH_3 has formed.

$N_2 + 3H_2 \rightleftharpoons 2NH_3$

Equilibrium moles: N_2 1.00 − 0.20 = 0.80; H_2 3.00 − 0.60 = 2.40; NH_3 0.40; **total 3.60**

Mole fractions: N_2 0.222; H_2 0.667; NH_3 0.111

Partial pressures (× 200): N_2 44.4 atm; H_2 133.3 atm; NH_3 22.2 atm

$K_p = 22.2^2 \div (44.4 \times 133.3^3) =$ **4.69×10^{-6} atm^{-2}**

WORKED EXAMPLE

Worked example 4: K_p for $N_2O_4 \rightleftharpoons 2NO_2$

1.00 mol N_2O_4 is 20% dissociated at 298 K and a total pressure of 1.00 atm.

Equilibrium moles: N_2O_4 0.80; NO_2 0.40; total 1.20

$p(N_2O_4) = (0.80/1.20) \times 1.00 = 0.667$ atm; $p(NO_2) = (0.40/1.20) \times 1.00 = 0.333$ atm

$K_p = 0.333^2 \div 0.667 =$ **0.167 atm**

WORKED EXAMPLE

Worked example 5: equilibrium amounts from K_c (no quadratic)

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$, $K_c = 49$ at a certain temperature. Start with 1.00 mol H_2 and 1.00 mol I_2. How much HI forms?

Equilibrium: H_2 (1.00 − x), I_2 (1.00 − x), HI 2x. K_c has no units, so volumes cancel.

$(2x)^2 \div (1.00 - x)^2 = 49 \rightarrow$ take the square root: $2x \div (1.00 - x) = 7$

$2x = 7 - 7x \rightarrow x = 0.778$ mol, so **HI = 1.56 mol**; $H_2 = I_2 = 0.222$ mol.

Step-through: 1: $H_2 + I_2 \rightleftharpoons 2HI$ (no units)

1. 0.500 mol of H_2 and 0.500 mol of I_2 are heated in a sealed 2.00 dm^3 flask. At equilibrium the flask contains 0.786 mol of HI. Find K_c.

2. HI formed = 0.786 mol. The equation says 2 HI form for every 1 H_2 and 1 I_2 used, so 0.393 mol of each reactant reacted.

3

	H_2	I_2	HI
Initial / mol	0.500	0.500	0
Change / mol	−0.393	−0.393	+0.786
Equilibrium / mol	0.107	0.107	0.786
Concentration / mol dm^{-3}	0.0535	0.0535	0.393

4 $K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{0.393^2}{0.0535 \times 0.0535} = 54.0$, with no units.

Because the units cancel, the volumes cancel too: putting the equilibrium moles straight in gives the same answer. That shortcut only works when K_c has no units.

Step-through: 2: $2SO_2 + O_2 \rightleftharpoons 2SO_3$ (with units)

1 2.00 mol of SO_2 and 1.20 mol of O_2 are mixed in a 5.00 dm^3 vessel with a catalyst. At equilibrium there are 1.60 mol of SO_3. Find K_c.

2 1.60 mol of SO_3 formed, so 1.60 mol of SO_2 and 0.800 mol of O_2 reacted (ratio 2 : 1 : 2).

3 Equilibrium moles: SO_2 2.00 − 1.60 = 0.400; O_2 1.20 − 0.800 = 0.400; SO_3 1.60.

4 Divide by 5.00 dm^3: $[SO_2]$ = 0.0800, $[O_2]$ = 0.0800, $[SO_3]$ = 0.320 mol dm^{-3}.

5 $K_c = \frac{[SO_3]^2}{[SO_2]^2[O_2]} = \frac{0.320^2}{0.0800^2 \times 0.0800} = 200\ \text{dm}^3\,\text{mol}^{-1}$

Here the volume does not cancel. Forgetting to divide by 5.00 would give 40.0, a wrong answer.

Step-through: 3: $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

1 1.00 mol of N_2O_4 is allowed to reach equilibrium. 20.0% of it dissociates, and the total pressure is 1.00 atm. Find K_p.

2. N_2O_4 reacted = 0.200 mol, so NO_2 formed = 0.400 mol. At equilibrium: 0.800 mol N_2O_4 and 0.400 mol NO_2, 1.200 mol in total.

3. Mole fractions: $x_{N_2O_4} = 0.800/1.200 = \frac{2}{3}$; $x_{NO_2} = 0.400/1.200 = \frac{1}{3}$.

4. Partial pressures at 1.00 atm: $p_{N_2O_4} = \frac{2}{3} = 0.667$ atm, $p_{NO_2} = \frac{1}{3} = 0.333$ atm.

5. $$K_p = \frac{p_{NO_2}^2}{p_{N_2O_4}} = \frac{(1/3)^2}{2/3} = 0.167 \text{ atm}$$

   With the rounded values 0.333 and 0.667 you would get 0.166: another reason to keep full figures until the end.

Step-through: 4: $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

1. 2.00 mol of N_2 and 6.00 mol of H_2 reach equilibrium at a total pressure of 100 atm. The equilibrium mixture contains 1.00 mol of NH_3. Find K_p.

2. 1.00 mol of NH_3 formed, so 0.500 mol of N_2 and 1.50 mol of H_2 reacted. At equilibrium: N_2 1.50, H_2 4.50, NH_3 1.00; total 7.00 mol.

3. Partial pressures: $p_{N_2} = \frac{1.50}{7.00} \times 100 = 21.4$ atm; $p_{H_2} = \frac{4.50}{7.00} \times 100 = 64.3$ atm; $p_{NH_3} = \frac{1.00}{7.00} \times 100 = 14.3$ atm. Check: they add up to 100 atm.

4. $$K_p = \frac{p_{NH_3}^2}{p_{N_2}\, p_{H_2}^3} = \frac{14.3^2}{21.4 \times 64.3^3} = 3.58 \times 10^{-5} \text{ atm}^{-2}$$

   Keep the unrounded partial pressures in your calculator; rounding to three figures first can shift the last digit.

Calculating Kc from the titration

1. The mixture started with 0.200 mol of ethanoic acid and 0.200 mol of ethanol. Each 25.0 cm^3 portion of the diluted mixture needed 21.40 cm^3 of 0.500 mol dm^{-3} NaOH. The catalyst alone contains 0.0400 mol of H^+.

2. NaOH used per portion = 0.02140 dm^3 × 0.500 mol dm^{-3} = 0.01070 mol, so the 25.0 cm^3 portion held 0.01070 mol of acid.

3. The whole 250 cm^3 held ten times as much: 0.1070 mol of acid.
4. Ethanoic acid at equilibrium = 0.1070 − 0.0400 = 0.0670 mol.
5. Ethanoic acid that reacted = 0.200 − 0.0670 = 0.133 mol, so 0.133 mol of ethanol reacted, and 0.133 mol each of ester and water formed. Ethanol left = 0.200 − 0.133 = 0.0670 mol.
6. $K_c = \frac{0.133 \times 0.133}{0.0670 \times 0.0670} = 3.94$, with no units, so the total volume cancels.

This ignores the water that came in with the dilute catalyst. Including it makes [H_2O] slightly bigger and K_c slightly bigger: a known limitation of the method.

MARK-SCHEME WORDING

- Typical mark pattern: equilibrium moles (1); mole fractions (1); partial pressures (1); substitution into K_p (1); value with units (1).
- Units: atm^{-1}, atm^{-2}, mol dm^{-3}, dm^3 mol^{-1}; "**no units**" when the powers cancel (WCH14 Oct 2022 Q18). SF: 2 or 3 usually.
- Ester equilibria: "a statement that the volumes cancel" earns the last mark (WCH14 Jan 2022 Q21, Jan 2024 Q19).

Often asked as: K_c from moles and a volume; K_p from moles → mole fractions → partial pressures; K_p from a percentage composition; show that a given mixture is not at equilibrium.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Converting K_p ↔ K_c.** For each gas, p = (n/V)RT = cRT, so $K_p = K_c(RT)^{\Delta n}$, where Δn = moles of gas products − moles of gas reactants. With p in atm and c in mol dm^{-3}, use **R = 0.0821 dm^3 atm K^{-1} mol^{-1}** and T in K (WCH14 Jan 2025 Q5, Oct 2025 Q14). For $N_2O_4 \rightleftharpoons 2NO_2$ (Δn = +1): $K_c = K_p \div RT = 0.167 \div (0.0821 \times 298) = \mathbf{6.8 \times 10^{-3}}$ **mol dm^{-3}**. For $CaCO_3 \rightleftharpoons CaO + CO_2$: $K_c = [CO_2] = p(CO_2) \div RT$.
- **pV = nRT** in SI units (p in Pa, V in **m^3**, R = 8.31 J K^{-1} mol^{-1}): e.g. 15 dm^3 = 0.015 m^3 (WCH14 Oct 2023 Q15).
- **The reaction quotient, Q**: same expression as K but with the concentrations or pressures you actually have. If Q = K the mixture is at equilibrium; if not, it isn't. Example: for $N_2O_4 \rightleftharpoons 2NO_2$ (K_p = 0.167 atm), a mixture with $p(NO_2) = p(N_2O_4) = 0.50$ atm has $Q_p = 0.50^2 \div 0.50 = 0.50$ atm > K_p, so it is **not** at equilibrium and will shift **left** (WCH14 Oct 2025 Q14: Q_p = 4.5 atm^{-1} against K_p = 35.2 atm^{-1}).
- Use the **total moles at equilibrium**, not at the start, for mole fractions (WCH14 Oct 2021 Q19 examiner report).

COMMON MISTAKE

Using initial moles in K. Using $M_r(O) = 16$ for O_2 (it's 32; about a third of candidates did this in one paper). Not dividing by the volume when K_c has units.

Rounding in the middle of the calculation; 4 or more SF; missing units.

SPEC 13.4 How temperature, pressure and a catalyst affect equilibrium composition

WHAT THIS SPEC POINT ASKS

Understand how, if at all, changing the temperature or pressure, or adding a catalyst, affects the equilibrium composition of a homogeneous or heterogeneous system.

Summary

Change	Effect on position	Effect on K
Increase temperature	shifts in the **endothermic** direction	**changes** (13.6)
Decrease temperature	shifts in the **exothermic** direction	**changes**
Increase pressure	shifts towards the side with **fewer moles of gas**	no change
Decrease pressure	shifts towards **more moles of gas**	no change
Add a catalyst	**no change**: forward and backward rates go up **equally**; equilibrium is just reached sooner	no change
Add more of a reactant	shifts to use it up (towards products)	no change

- In a **heterogeneous** system, count **only gas moles** when you think about pressure. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$: 0 → 1 mol gas, so high pressure shifts it left. Adding more solid $CaCO_3$ doesn't shift the position.
- **Equal moles of gas on both sides** ($H_2 + I_2 \rightleftharpoons 2HI$): pressure doesn't shift the position.

Observations with coloured gases

- $N_2O_4(g,$ colourless$) \rightleftharpoons 2NO_2(g,$ brown$)$, forward endothermic.

- **Compress** it: the colour **darkens at once** (NO_2 is squeezed into a smaller volume), then **fades a little** as the position shifts left (fewer moles of gas). It stays darker than at the start.
- **Heat** it: browner, because the endothermic forward reaction is favoured.
- For $H_2 + I_2 \rightleftharpoons 2HI$, compressing makes the purple colour more intense but **doesn't** shift the position.

Industrial compromise: the Haber process

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), \Delta H = -92 \text{ kJ mol}^{-1}$$

- **Temperature about 450 °C**: a low temperature would give a higher yield (exothermic forward), but the rate would be too slow. 450 °C gives a reasonable yield at a fast enough rate.
- **Pressure about 200 atm**: high pressure favours NH_3 (4 → 2 mol gas) and increases the rate, but very high pressures cost too much to build and run safely.
- **Iron catalyst**: faster, no effect on yield.
- Only about 15% conversion per pass: unreacted gases are **recycled**, so the overall conversion is much higher.

MARK-SCHEME WORDING

- "increase in temperature would decrease yield / shift the equilibrium to the left, because the reverse reaction is endothermic"; "increase in pressure would increase the yield because there are fewer moles (of gas) on the right-hand side"; "a catalyst has no effect on the yield because the rates of the forward and backward reactions are increased (equally)" (WCH14 Jan 2026 Q15). "Molecules" for "moles" is allowed.
- "when the pressure is increased, equilibrium moves to the RHS and yield increases as fewer gas molecules on the RHS; **K_p remains constant**"; "temperature increase: equilibrium moves to LHS as the forward reaction is exothermic; **K_p decreases**" (WCH14 Jun 2021 Q20).
- Large K: "the equilibrium is (already) far to the right, so increased pressure would give only a small increase in yield" (WCH14 Oct 2025 Q14).

Often asked as: effect of each change on yield and on K, with reasons; colour change on compressing; justify compromise conditions.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Compressing an equilibrium with **equal gas moles** on both sides: the position doesn't move, but the colour gets **more intense** because the same particles are in a smaller volume (WCH14 Jan 2025 Q10, the hardest MCQ on that paper).
- **Recycling** unreacted gases to get overall conversion above the equilibrium yield (WCH14 Oct 2021 Q19, Jan 2022 Q22).

COMMON MISTAKE

Saying "the equilibrium shifts left" without saying what happens to the **yield**.

Mixing up exothermic and endothermic directions.

Using Le Chatelier when the question says "in terms of K_p": then you must argue with Q and K (13.5).

SPEC 13.5 K doesn't change with concentration, pressure or a catalyst

WHAT THIS SPEC POINT ASKS

Understand that the value of the equilibrium constant is not affected by changes in concentration or pressure, or by adding a catalyst.

Only temperature changes K

- **Concentration**: adding a reactant makes the quotient Q smaller than K. The system reacts forwards (making more products, using reactants) until **Q = K** again. The position moves; K doesn't.
- **Pressure**: changing the total pressure changes every partial pressure by the same factor. If the gas moles differ on each side, Q changes; the system shifts until Q = K_p again.
- **Catalyst**: speeds up forward and reverse reactions by the same factor. Equilibrium arrives sooner with the same composition, so K is unchanged.

WORKED EXAMPLE

Worked example: pressure explained with K_p

N_2 + 3H_2 ⇌ 2NH_3: double the total pressure. Every partial pressure doubles at once.

$Q_p = (2p(NH_3))^2 \div ((2p(N_2)) \times (2p(H_2))^3) = (4 \div 16) \times K_p =$ **$K_p \div 4$**

Q_p is now smaller than K_p, so the system makes more NH_3 (top line up, bottom line down) until $Q_p = K_p$ again. **Position shifts right; K_p unchanged.**

N_2O_4 ⇌ 2NO_2: double the pressure.

$Q_p = (2p(NO_2))^2 \div (2p(N_2O_4)) =$ **$2 \times K_p$**

Q_p is too big, so NO_2 turns back into N_2O_4 until $Q_p = K_p$. **Position shifts left.**

Why higher pressure gives more ammonia although Kp stays the same

1. Write each partial pressure as mole fraction × total pressure: $K_p = \dfrac{(x_{NH_3}P)^2}{(x_{N_2}P)(x_{H_2}P)^3}$.

2. Collect the P terms: $K_p = \dfrac{x_{NH_3}^2}{x_{N_2}\,x_{H_2}^3} \times \dfrac{1}{P^2}$.

3. Double P and the factor $1/P^2$ falls to a quarter. Straight after the change the right-hand side is smaller than K_p, so the mixture is not at equilibrium.

4. To restore K_p, the mole-fraction part must grow fourfold: more NH_3 forms and N_2 and H_2 are used up. The position shifts right, towards fewer gas molecules, and K_p ends where it started.

If both sides have the same number of gas molecules, the P terms cancel completely and pressure has no effect on the position at all.

MARK-SCHEME WORDING

- "the value of K_p would not change due to increased pressure or the presence of a catalyst, (but) temperature changes K_p" (WCH14 Jan 2026 Q15).
- Increased pressure "in terms of K_p": "K_p remains the same"; "the quotient / Q_p / apparent value of K_p decreases (when pressure increases)"; "so equilibrium shifts to the right-hand side (to return to the value of K_p)" (WCH14 Oct 2020 Q19).

Often asked as: state and explain what happens to K; explain a pressure change using K_p.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The **reaction quotient Q** argument is the only route credited when a question says "in terms of K_p" (WCH14 Oct 2020 Q19). The same idea shows whether a mixture is at equilibrium (WCH14 Oct 2025 Q14).

COMMON MISTAKE

Saying K increases because "more product is made at higher pressure". K is fixed; the position moves to keep it fixed.

Leaving out what happens to the value of K when asked (examiner report, Jan 2021).

SPEC 13.6 How temperature changes K: exothermic and endothermic reactions

WHAT THIS SPEC POINT ASKS

Know the effect of changing temperature on K_c and K_p for exothermic and endothermic reactions.

Forward reaction	Increase T	Decrease T
Exothermic ($\Delta H < 0$), e.g. $2SO_2 + O_2 \rightleftharpoons 2SO_3$; $N_2 + 3H_2 \rightleftharpoons 2NH_3$	**K decreases**: less product	K increases
Endothermic ($\Delta H > 0$), e.g. $N_2O_4 \rightleftharpoons 2NO_2$; $CaCO_3 \rightleftharpoons CaO + CO_2$	**K increases**: more product	K decreases

Memory aid: heating always favours the **endothermic** direction. If that's the forward direction, K goes up.

WORKED EXAMPLE

Worked example: reading ΔH from K data

For $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, K_p is 3.4 atm^{-1} at 1000 K and 0.13 atm^{-1} at 1200 K (illustrative values).

K_p falls as T rises → less SO_3 at equilibrium → the forward reaction must be **exothermic**.

In terms of the expression: the partial pressure of SO_3 (top) has decreased and those of SO_2 and O_2 (bottom) have increased.

MARK-SCHEME WORDING

- "(decreasing K_p means) the partial pressure of products decreases; the partial pressure of reactants increases; so equilibrium moves to the left (at higher temperature)"; "numerator decreases / denominator increases" allowed (WCH14 Oct 2025 Q14).
- "(forward) reaction is endothermic; at the lower temperature the reaction moves in the exothermic direction / backwards (making K_c smaller)" (WCH14 Jun 2025 Q14).
- "the equilibrium constant decreases (as temperature increases) because the forward reaction is exothermic". Both the yield and the change in K are needed (WCH14 Jan 2021 Q18).

Often asked as: predict the change in K; deduce the sign of ΔH from K at two temperatures; MCQ "which change alone increases K".

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Choosing the **one** change that increases K from a list (pressure, catalyst, heating, cooling): only a temperature change can, and which one depends on the sign of ΔH (WCH14 Jan 2025 Q10).

COMMON MISTAKE

Saying a catalyst or pressure changes K. Saying "K shifts to the right": K is a number; it increases or decreases.

SPEC 13.7 Temperature moves the equilibrium position by changing K

WHAT THIS SPEC POINT ASKS

Understand that the effect of temperature on the position of equilibrium is explained by a change in the value of the equilibrium constant.

The chain of reasoning, for an **exothermic** forward reaction heated up:

1. Raising the temperature makes **K smaller** (13.6, explained by entropy in 13.8).
2. At that instant the concentrations haven't changed, so the quotient Q is still the old value: now **Q > K**.
3. To make Q fall to the new K, products must turn back into reactants.
4. So the position shifts **left** and the yield falls.

For an endothermic forward reaction, heating makes K bigger, so Q < K and the position shifts right.

Compare with pressure and concentration: there **Q changes and K stays**; with temperature **K changes and Q has to follow**.

MARK-SCHEME WORDING

- "temperature changes K" is the reason the position shifts (WCH14 Jan 2026 Q15).
- Linking to entropy: "$\Delta_r H$ is negative (as increasing temperature moves the equilibrium to the left) … $\Delta S_{surroundings}$ becomes less positive as T increases … ΔS_{total} has decreased" (WCH14 Oct 2025 Q14).

Often asked as: explain in terms of K why heating changes the yield.

COMMON MISTAKE

Explaining a temperature change with Le Chatelier alone when the question asks for K.

SPEC 13.8 $\Delta S_{total} = R \ln K$: how temperature affects ΔS_{total} and K

WHAT THIS SPEC POINT ASKS

Understand the effect of a change in temperature on (i) ΔS_{total} and (ii) the size of the equilibrium constant, since $\Delta S_{total} = R \ln K$.

The link

$\Delta S_{total} = R \ln K$, so **$\ln K = \Delta S_{total} \div R$** and **$K = e^{\Delta S_{total}/R}$** ($R = 8.31$ J K^{-1} mol^{-1}, ΔS_{total} in J K^{-1} mol^{-1}, standard values). You have to remember this: it isn't given.

- ΔS_{total} large and positive → K very large → products dominate.
- $\Delta S_{total} = 0$ → $K = 1$.
- ΔS_{total} negative → $K < 1$ → reactants dominate.

Temperature: the full chain

$\Delta S_{total} = \Delta S_{system} - \Delta H/T$. ΔS_{system} and ΔH barely change with temperature, so only the $-\Delta H/T$ term changes.

Exothermic forward reaction ($\Delta H < 0$):

increase T → $\Delta S_{surroundings}$ (= $-\Delta H/T$, positive) gets **smaller** → ΔS_{total} **decreases** → ln K decreases → **K decreases**.

Endothermic forward reaction ($\Delta H > 0$):

increase T → $\Delta S_{surroundings}$ (negative) gets **less negative** → ΔS_{total} **increases** → **K increases**.

If $\Delta H \approx 0$, $\Delta S_{surroundings}$ hardly changes with T, so K hardly changes.

WORKED EXAMPLE

Worked example: K for the Haber process at two temperatures

$N_2 + 3H_2 \rightleftharpoons 2NH_3$; $\Delta H = -92.2$ kJ mol^{-1}; $\Delta S_{system} = -198.8$ J K^{-1} mol^{-1}

298 K: $\Delta S_{total} = -198.8 + 92\,200/298 = +110.6$ J K^{-1} mol^{-1}

ln K = 110.6 ÷ 8.31 = 13.31, so **$K = 6.0 \times 10^5$**: equilibrium lies far to the right.

700 K: ΔS_{total} = −198.8 + 92 200/700 = −198.8 + 131.7 = −67.1 J K^{-1} mol^{-1}

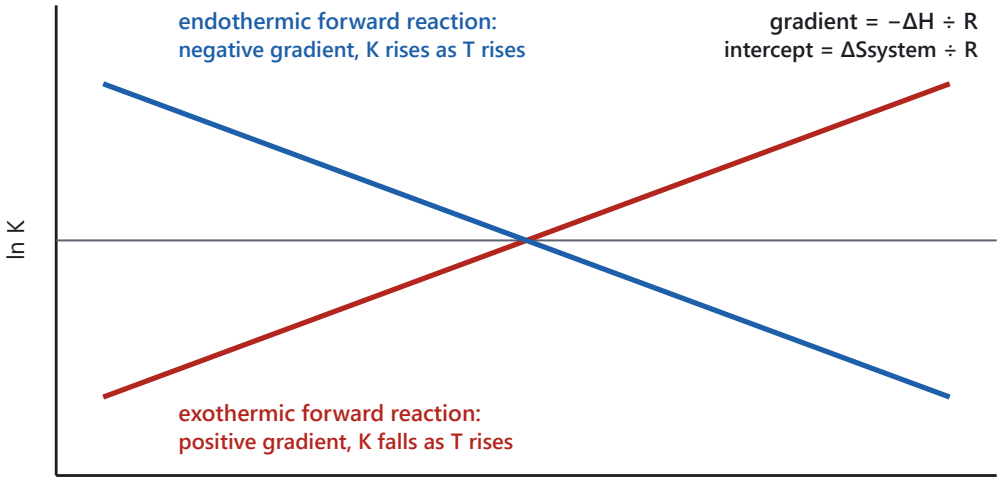
ln K = −67.1 ÷ 8.31 = −8.07, so **K = 3.1 × 10^{-4}**: the yield at equilibrium is low.

That's why the plant needs high pressure and recycling to compensate for the high temperature it uses for rate.

WORKED EXAMPLE

Worked example: ΔS_{total} from K

K = 0.484 at 298 K. ΔS_{total} = R ln K = 8.31 × ln 0.484 = 8.31 × (−0.726) = **−6.03 J K^{-1} mol^{-1}** (K < 1, so ΔS_{total} is negative).



ln K = $\Delta S_{system}/R$ − (ΔH/R)(1/T). The gradient is −ΔH/R, so an exothermic reaction gives a positive gradient.

Why temperature changes K

1. ΔS_{total} = ΔS_{system} + $\Delta S_{surroundings}$, and $\Delta S_{\text{surroundings}} = -\frac{\Delta H}{T}$.
2. For an **exothermic** reaction ΔH is negative, so $\Delta S_{surroundings}$ is positive. As T rises, −ΔH/T gets **smaller**.
3. So ΔS_{total} becomes less positive (or more negative). Because ΔS_{total} = R ln K, ln K falls and K gets smaller: less product at equilibrium.

4 For an **endothermic** reaction $\Delta S_{surroundings}$ is negative, and rising T makes it less negative. ΔS_{total} increases, so K gets larger.

This is the quantitative reason behind the Le Chatelier rule you learned at AS: heating favours the endothermic direction.

Gibbs energy and the equilibrium constant

$$\Delta G = -RT \ln K$$

R = 8.31 J K⁻¹ mol⁻¹, so ΔG here is in J mol⁻¹ T in K

MARK-SCHEME WORDING

- "ΔS_{total} = R ln K" recalled; "ln K = ΔS_{total} ÷ R"; e.g. ln K = 1780.7 ÷ 8.31 = 214.2, K = 1.15 × 10^{93} (WCH14 Jan 2020 Q19). No TE from a wrong expression.
- Higher T, exothermic: "ΔS_{system} and ΔH barely change with T, so $\Delta S_{surroundings}$ = −ΔH/T becomes less positive, so ΔS_{total} decreases" (WCH14 Jan 2022 Q22).
- ΔH close to zero: "$\Delta S_{surroundings}$ / ΔS_{total} only changes by a small amount with temperature", so "very little change in the equilibrium constant" (WCH14 Jan 2024 Q19).

Often asked as: calculate K from ΔS_{total} (or the reverse); explain how K changes with temperature using ΔS_{total}; MCQ on the relationship.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **ln K against 1/T** (the van 't Hoff plot): ln K = −(ΔH/R)(1/T) + ΔS_{system}/R, so **gradient = −ΔH/R** and **intercept = ΔS_{system}/R** (WCH14 Oct 2022 Q18). Same shape of argument as ΔS_{total} against 1/T in 12.8.
- **ΔG = −RT ln K**, and ΔS_{system} = (ΔH − ΔG) ÷ T (WCH14 Oct 2022 Q18).

WORKED EXAMPLE

Worked example: ΔH and ΔS_{system} from a ln K against 1/T graph

The gradient of ln K against 1/T is +5000 K, and K = 50 at 500 K.

$\Delta H = -\text{gradient} \times R = -5000 \times 8.31 = -41\,600$ J mol^{-1} = **−41.6 kJ mol^{-1}** (positive gradient → exothermic)

$\Delta G = -RT \ln K = -8.31 \times 500 \times \ln 50 = -8.31 \times 500 \times 3.91 = -16\,300$ J mol^{-1}

$\Delta S_{system} = (\Delta H - \Delta G) \div T = (-41\,600 - (-16\,300)) \div 500$ = **−50.6 J K^{-1} mol^{-1}**

COMMON MISTAKE

Using kJ for ΔH in $\Delta S_{total} = \Delta S_{system} - \Delta H/T$ (it must be J). Forgetting to take e^x to get from ln K to K.

Units of ΔG: J **mol^{-1}**, not J.

Describing $\Delta S_{surroundings}$ generically when the question is about a reaction with ΔH close to zero.

SPEC 13.9 Using K to judge how far a reaction goes

WHAT THIS SPEC POINT ASKS

Use the value of an equilibrium constant to predict the extent to which a reaction takes place.

Size of K	ΔS_{total} (= R ln K)	What's in the equilibrium mixture
K > about 10^{10}	more than about +200 J K^{-1} mol^{-1}	**essentially all products**: reaction "goes to completion"
K ≈ 1	≈ 0	**significant amounts of both** reactants and products
K < about 10^{-10}	below about −200 J K^{-1} mol^{-1}	**essentially no products**: reaction "doesn't go"

- A **large** K means the equilibrium lies **far to the right**; a small K means it lies far to the left.
- K tells you **how far**, never **how fast**. A reaction with K = 10^{50} can still be too slow to see if E_a is high (12.11).
- Where K is already huge, changing the pressure or concentration gains very little extra yield.

MARK-SCHEME WORDING

- "the equilibrium constant is (very) large"; "the equilibrium position is (far) to the right / (heavily) favours the products". "$K_p \gg 1$", "huge", "massive" allowed. IGNORE "quite large", "$K_p > 1$", "positive" (WCH14 Jun 2023 Q18).
- K < 1: reactants favoured (WCH14 Jun 2024 Q5).

Often asked as: what a very large K tells you about a reaction (2 marks); MCQ on what K < 1 means; why high pressure isn't worth it when K is already large.

COMMON MISTAKE

Saying "K is greater than 1" instead of commenting on how **large** it is. The mean mark on one 2-mark question was below 1 for this reason (examiner report, Jun 2023).

Describing the equilibrium "moving to the right" when the question is about the **final position**.

Assuming a large K means a fast reaction.

Quick checks

QUICK CHECK

What are the units of K_c for $2NO_2(g) \rightleftharpoons N_2O_4(g)$?

- mol dm^{-3}
- dm^3 mol^{-1}
- no units
- mol^2 dm^{-6}

$K_c = [N_2O_4]/[NO_2]^2$: one concentration on top and two underneath gives (mol dm^{-3})$^{-1}$ = dm^3 mol^{-1}. mol dm^{-3} is the unit for the reverse equation.

QUICK CHECK

Which is the correct K_p expression for $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$?

- $\dfrac{p_{CO}\, p_{H_2}}{p_C\, p_{H_2O}}$
- $\dfrac{p_{H_2O}}{p_{CO}\, p_{H_2}}$
- ✓ $\dfrac{p_{CO}\, p_{H_2}}{p_{H_2O}}$
- $p_{CO}\, p_{H_2}$

Carbon is a solid, so it does not appear; steam is a gas and does. Products go on top.

QUICK CHECK

A gas mixture contains 0.60 mol of A and 1.40 mol of B at a total pressure of 250 kPa. What is the partial pressure of A?

- ✓ 75 kPa
- 107 kPa
- 150 kPa
- 0.30 kPa

$x_A = 0.60 \div 2.00 = 0.30$, so $p_A = 0.30 \times 250 = 75$ kPa. 107 kPa comes from dividing by the moles of B instead of the total.

QUICK CHECK

For an exothermic equilibrium, what happens when the temperature is raised?

- K increases and the position moves right
- K is unchanged but the position moves left
- K decreases but the position does not change
- K decreases and the position moves left

Temperature is the only thing that changes K. For an exothermic reaction $\Delta S_{surroundings} = -\Delta H/T$ falls as T rises, so ΔS_{total} and K fall, and the mixture holds less product.

QUICK CHECK

The pressure on the equilibrium $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is increased at constant temperature. What happens to K_p and to the amount of SO_3?

- K_p increases; more SO_3
- K_p unchanged; more SO_3
- K_p unchanged; less SO_3
- K_p decreases; more SO_3

K_p depends only on temperature. With 3 gas molecules on the left and 2 on the right, higher pressure shifts the position right, so the amount of SO_3 rises.

QUICK CHECK

A reaction has $K_c = 2 \times 10^{12}$ at 298 K. What can you say about it?

- It is fast
- Reactants predominate at equilibrium
- It goes essentially to completion, and ΔS_{total} is positive
- ΔS_{total} is negative

K above about 10^{10} means essentially complete reaction, and $\Delta S_{total} = R \ln K$ is positive. K says nothing about the rate: a reaction can have a huge K and still be very slow.

QUICK CHECK

0.40 mol of PCl_5 is heated in a 2.0 dm^3 flask: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$. At equilibrium 0.10 mol of Cl_2 is present. What is K_c?

- 0.033 mol dm^{-3}
- 0.067 mol dm^{-3}
- 0.025 mol dm^{-3}
- 0.017 mol dm^{-3}

Equilibrium: PCl_5 0.30 mol, PCl_3 0.10 mol, Cl_2 0.10 mol. Concentrations 0.15, 0.050 and 0.050 mol dm^{-3}, so $K_c = (0.050 \times 0.050) \div 0.15 = 0.017$ mol dm^{-3}. Using moles without dividing by the volume gives 0.033.

QUICK CHECK

Why is the Haber process run at about 450 °C rather than at a lower temperature?

- K is larger at 450 °C
- A lower temperature would give a better yield but far too slow a rate
- The iron catalyst increases K at 450 °C
- Ammonia decomposes below 450 °C

The forward reaction is exothermic, so K is actually larger at lower temperatures. 450 °C is a compromise that gives an acceptable rate with a reasonable yield. A catalyst never changes K.